

Important Topics to Practice for Midterm

1. 2's complement operation in given bits and overflow detection

As given in Assignment 2 & Quiz 2.

2. Boolean to Canonical SOP (Minterms), POS* (Maxterms)

Some practice problems:

a. $x+y(x'+y'z)+yz$

[POS DETAILED SOLUTION](#) (try yourself before checking the solution out)

b. $xz+y'x+z(xz'+y)$

3. Simplifying boolean functions using laws of Boolean algebra only

Some practice problems:

a. $F(a,b,c) = (a'+b'+c)(a+b+c')(b'+c')$

b. $F(x,y,z) = xyz + x'(y'z+x)' + yz'(y+x')$

c. $F(a,b,c) = c(ab+bc')' + a'b(c+b'c) + a'bc$

4. Implementing functions using NAND/NOR

Try implementing the simplified functions from Q3 using NAND/NOR gates.

Remember the steps (refer to [Lecture 3](#), slide #38-41):

1. Draw the function using basic gates first
2. Replace the basic gates with their NAND/NOR equivalent circuits
3. Remove two (or even number of) cascading (in series) NOT gates as they cancel each other.

5. K-Map Word Problems (4 variables) with Multiple Outputs.

Solve the one given in this doc and also practice the ones in your buX practice [sheet](#).

Statement Problem Practice:

BracU is going to introduce two types of courses for the Spring 23 semester, online & offline. Design a system that will determine the eligibility of a student to do an online or an offline course based on the following criteria:

Inputs

1. **U:** Current Location (U = 1: student is at university, U = 0: at home)
2. **L:** Laptop (L = 1, student has a laptop, P = 0 means otherwise)
3. **H:** Headphones (P = 1: has headphones, P = 0 means otherwise)
4. **V:** Vaccine (V = 1: student is vaccinated, V = 0 means otherwise)

Outputs

A: Offline Course

B: Online Course

The outputs will depend on the following conditions:

- Offline courses can be taken if the student is present in the university and either s/he doesn't have a laptop or headphones.
- Students must be vaccinated to attend an offline course.
- For an online course, if at home the student must have both a laptop and headphones.
- Students can also attend an online class from university if he is vaccinated, or he has a laptop with headphones
- The system can suggest both online & offline classes.

a) Devise a truth table for the system above which will suggest a type of course based on the given inputs. In the truth table, maintain the ULHV sequence, U is the MSB, V is the LSB.

b) Use K-Maps to derive simplified equations for both of the outputs.

Quiz 2 Questions

- C01** Q1. Answer the following questions. 3
- a. **Represent** +127 & -128 in 10-bit 2's complement form
 - b. **Subtract** -128 from +127 in 10-bit 2's complement. State if there is an overflow.
- C01** Q2. **Simplify** the following expressions to the minimum number of literals using the laws of Boolean algebra. 3
- a. $x'y'z + x'yz + xyz'$
 - b. $(x+y')(x'+y)$
- C01** Q3. **Draw** the function $F = AB + A'C$ using: 4
- a. NAND gates only.
 - b. NOR gates only.
- C01** Q4. **Convert** $F = AB' + B'C$ to: 5
- a. Canonical SOP & write the Minterms.
 - b. Canonical POS & write the Maxterms.

